CHAPTER 5 - Discrete Probability Distributions

SDS188 - Group 3

2026

1 Expected value and variance

$$E(X) = \sum_{i=1}^{n} XP(X)$$

X	$P(X)$	$X \times P(X)$
x_1	$P(X = x_1)$	$x_1 \times P(X = x_1)$
x_2	$P(X = x_2)$	$x_2 \times P(X = x_2)$
$\vdots$	$\vdots$	$\vdots$
		$\sum_{i=1}^{n} x_i P(X = x_i)$

$$Var(X) = \sum_{i=1}^{n} (x_i - E(X))^2 P(X = x_i)$$

OF

$$Var(X) = E(X^2) - [E(X)]^2$$

X	$P(X)$	$X \times P(X)$	X^2	$X^2 \times P(X)$
x_1	$P(X = x_1)$	$x_1 \times P(X = x_1)$	x_1^2	$x_1^2 \times P(X = x_1)$
x_2	$P(X = x_2)$	$x_2 \times P(X = x_2)$	x_2	$x_2^2 \times P(X = x_2)$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
		$\sum_{i=1}^{n} XP(X)$		$\sum_{i=1}^{n} x_i^2 P(X = x_i)$

$$E(X^2) = x_i^2 P(X = x_i)$$

$$Var(X) = \boxed{E(X^2)} - [\boxed{E(X)}]^2$$

1.1 Example 1

The following table contains the probability distribution for the number of car accidents per day in a small town. What is the average and standard deviation of the number of accidents per day?
X = number of car accidents per day in a small town
$x = 0, 1, 2, 3, 4, 5$

X	$P(X)$	$X \times P(X)$	$X^2 \times P(X)$
0	0.1		
1	0.2		
2	0.45		
3	0.15		
4	0.05		
5	0.05		

1.2 Example 2

The manager of the mortgage loan department at a bank collected data for the past two years regarding the number of mortgage loans approved per week. The results from these two years (104 weeks) are as follows.
X = number of mortgage loans approved per week
$x = 0, 1, 2, 3, 4, 5, 6, 7$

X	F	$P(X)$	$X \times P(X)$	$X^2 \times P(X)$
0	13			
1	25			
2	32			
3	17			
4	9			
5	6			
6	1			
7	1			
	104			

1.3 Proofs

$$\begin{aligned}
E(aX+b) &= \sum_{i=1}^{N}(ax_i+b)P(X=x_i) \\
&= \sum_{i=1}^{N} ax_i P(X=x_i) + \sum_{i=1}^{N} bP(X=x_i) \\
&= a\sum_{i=1}^{N} x_i P(X=x_i) + b\sum_{i=1}^{N} P(X=x_i) \\
&= aE(X) + b(1) \\
&= a\mu_X + b
\end{aligned}$$

$$
\begin{aligned}
[E(aX+b)]^2 &= (aX+b)^2 \\
&= (aX+b)(aX+b) \\
&= a^2\mu_X^2 + a\mu_X b + ba\mu_X + b^2 \\
&= a^2\mu_X^2 + 2ab\mu_X + b^2
\end{aligned}
$$

$$
\begin{aligned}
E[(aX+b)^2] &= E[(aX+b)(aX+b)] \\
&= E(a^2X^2 + aXb + baX + b^2) \\
&= E(a^2X^2 + 2abX + b^2) \\
&= E(a^2X^2) + E(2abX) + E(b^2) \\
&= a^2E(X^2) + 2abE(X) + b^2 \\
&= a^2E(X^2) + 2ab\mu_X + b^2
\end{aligned}
$$

$$
\begin{aligned}
Var(aX+b) &= E[(aX+b)^2] - [E(aX+b)]^2 \\
&= a^2E(X^2) + 2ab\mu_X + b^2 - (a^2\mu_X^2 + 2ab\mu_X + b^2) \\
&= a^2E(X^2) + 2ab\mu_X + b^2 - a^2\mu_X^2 - 2ab\mu_X - b^2 \\
&= a^2[E(X^2) - \mu_X^2] + \cancel{2ab\mu_X} - \cancel{2ab\mu_X} + \cancel{b^2} - \cancel{b^2} \\
&= a^2 Var(X)
\end{aligned}
$$

2 Cumulative probabilities

$$
\begin{aligned}
P(X \le x) &= P(X = x) + P(X = (x-1)) + P(X = (x-2)) + \cdots + P(X = 0) \\
P(X \ge x) &= 1 - P(X \le (x-1)) = P(X = x) + P(X = (x+1)) + \cdots + P(X = n)
\end{aligned}
$$

3 Binomial distribution

Type of variable: Discrete
X = number of successes out of n observations
$x = 0, 1, 2, ..., n$

3.1 Parameters

n = number of observations
π = probability of success

3.2 Probability

If $X\ binomial(n, \pi)$, what is the probability that X equals x?
OR
What is the probability of x sucesses out of n observations?

$$P(X = x|n, \pi) = \frac{n!}{x!(n-x)!}\pi^x(1-\pi)^(n-x)$$

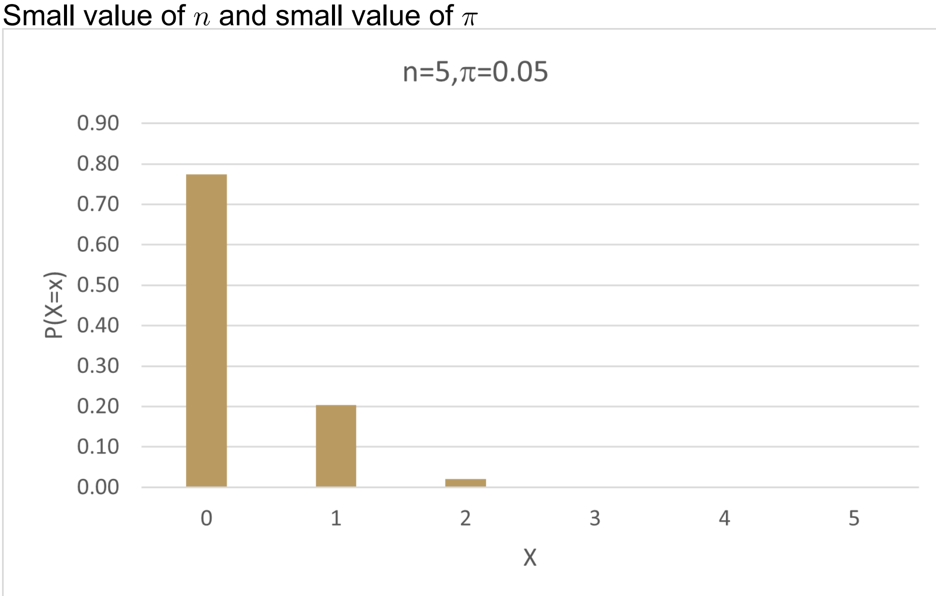
$${}_nC_x = \binom{n}{x} = \frac{n!}{x!(n-x)!}$$

X	$P(X = x\|n, \pi)$
0	$\frac{n!}{0!(n-0)!}\pi^0(1-\pi)^{n-0}$
1	$\frac{n!}{1!(n-1)!}\pi^1(1-\pi)^{n-1}$
$\vdots$	$\vdots$
n	$\frac{n!}{n!(n-n)!}\pi^n(1-\pi)^{n-n}$

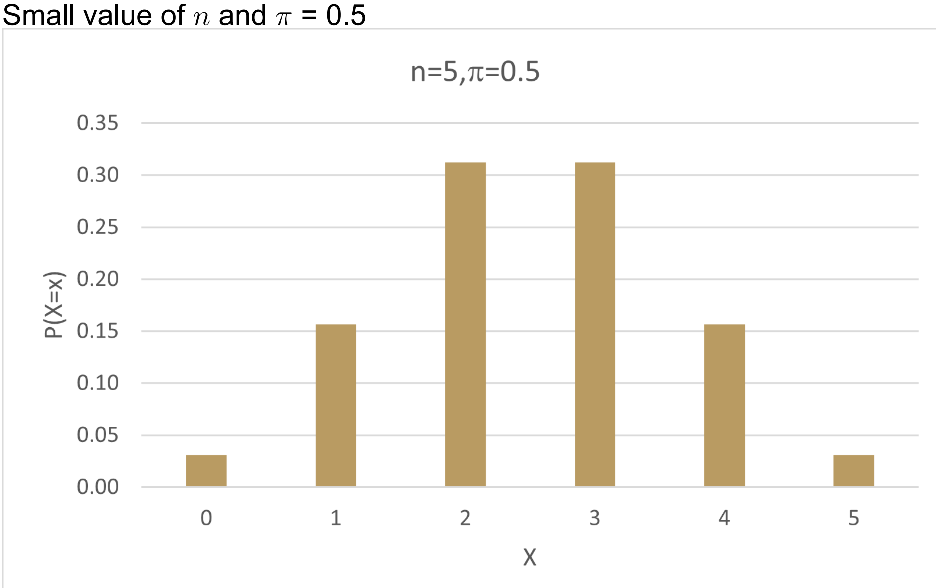
In Excel:
$P(X = x)$ =BINOM.DIST(x,n,π,FALSE)
$P(X \leq x)$ =BINOM.DIST(x,n,π,TRUE)

3.3 Effect of parameters on shape of distribution

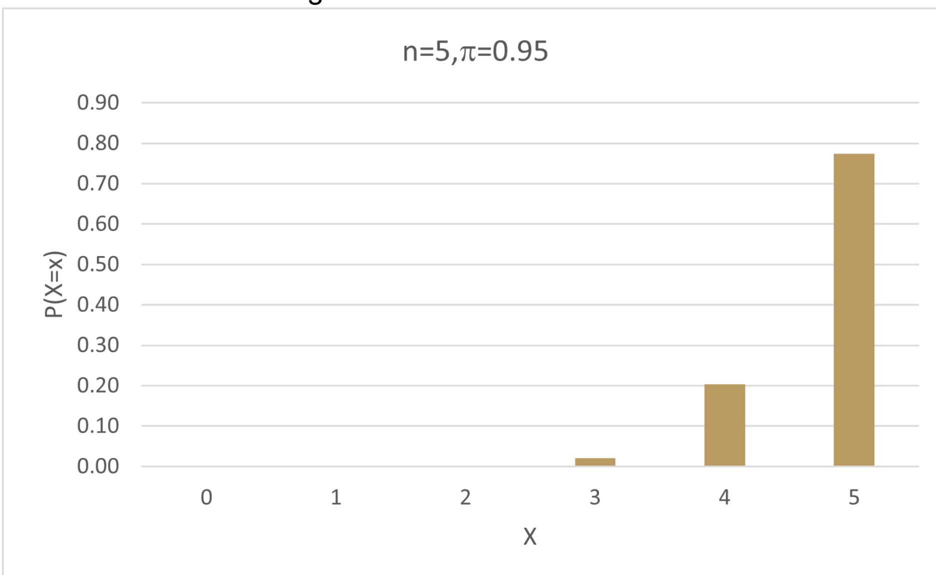
- Small value of n and small value of π



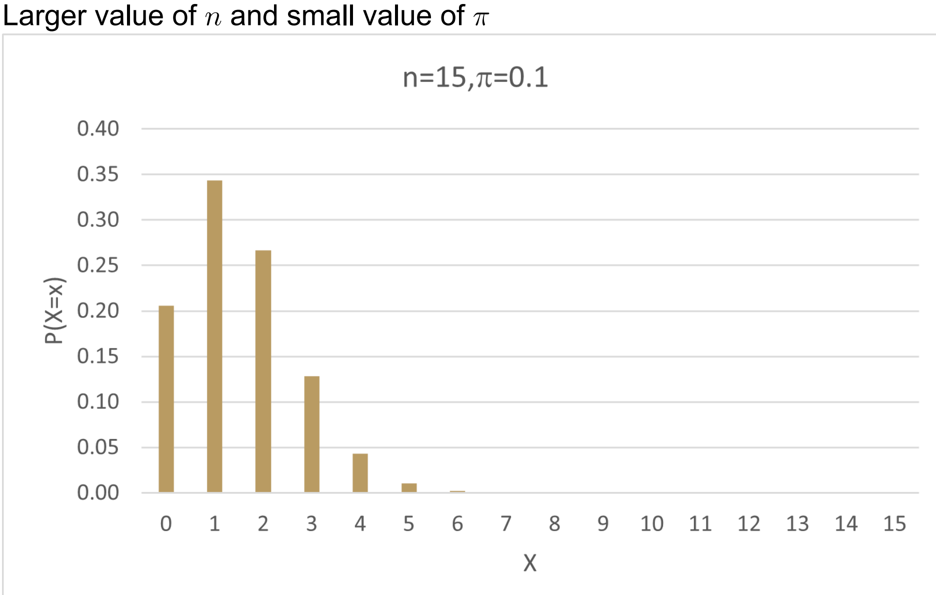
- Small value of n and π = 0.5



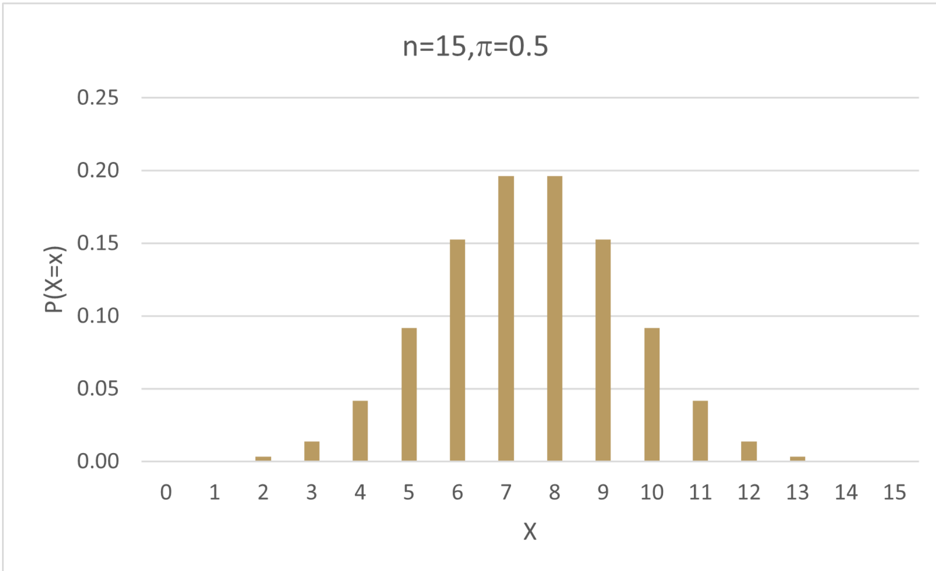
- Small value of n and large value of π



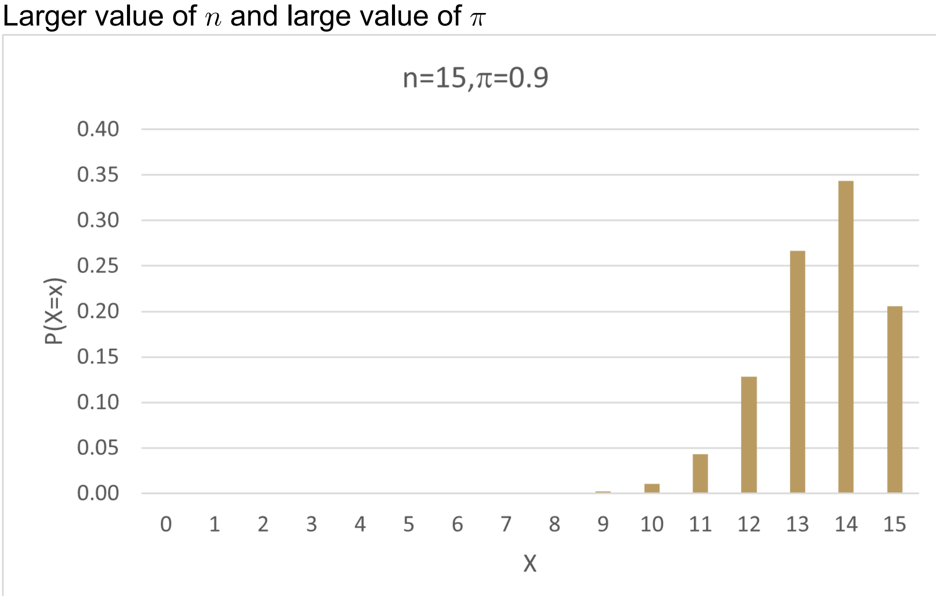
- Larger value of n and small value of π



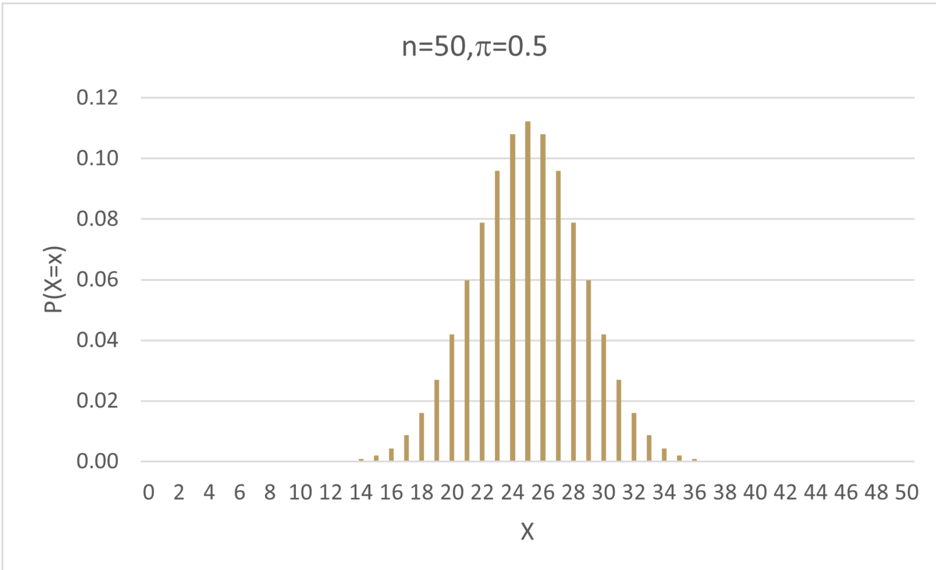
- Larger value of n and π = 0.5



- Larger value of n and large value of π



Larger values of n leads to a more symmetric distribution.



3.4 Measures

$$\mu = E(X) = n\pi$$
$$\sigma^2 = Var(X) = n\pi(1-\pi)$$

4 Poisson distribution

Type of variable: Discrete
X = number of events per area of opportunity
x = 0,1,2,...

4.1 Parameters

λ = expected number of events per area of opportunity

4.2 Probability

If X $poisson(\lambda)$, what is the probability that x events occur per area of opportunity?

$$P(X = x|\lambda) = \frac{e^{-\lambda}\lambda^x}{x!}$$

X	$P(X = x\|\lambda)$
0	$\frac{e^{-\lambda}\lambda^0}{0!}$
1	$\frac{e^{-\lambda}\lambda^1}{1!}$
$\vdots$	$\vdots$
n	$\frac{e^{-\lambda}\lambda^n}{n!}$

In **Excel**:

$P(X = x)$ =POISSON.DIST(x,λ,FALSE)

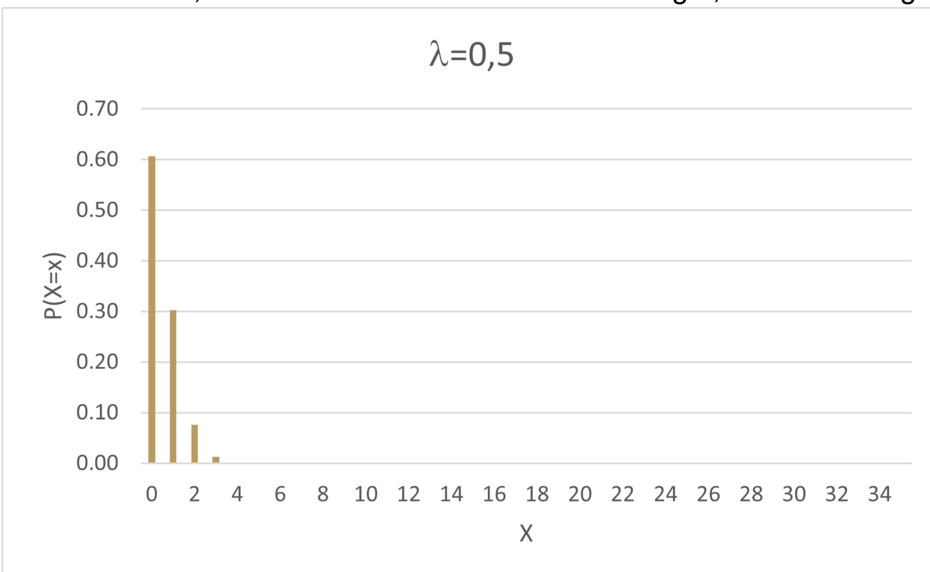
$P(X \leq x)$ =POISSON.DIST(x,λ,TRUE)

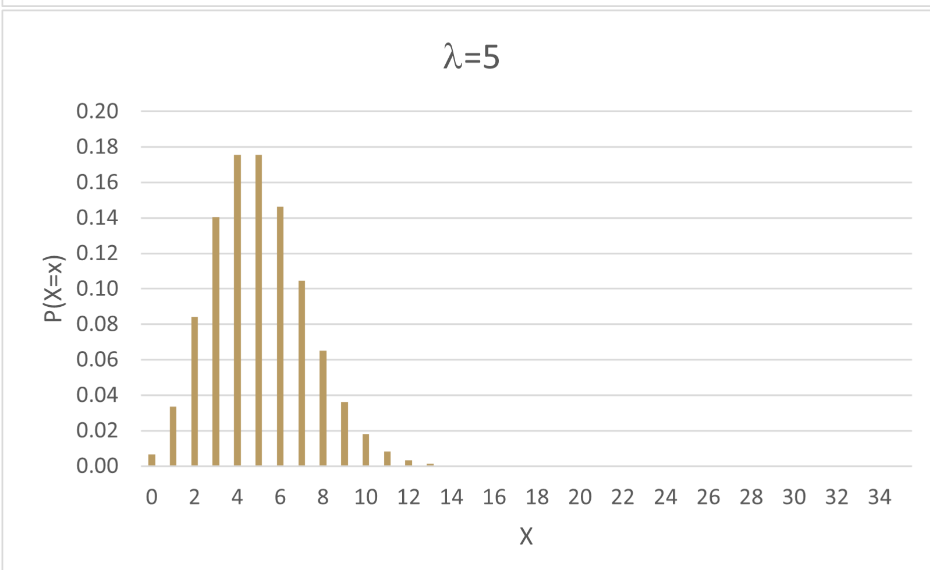
4.3 Measure

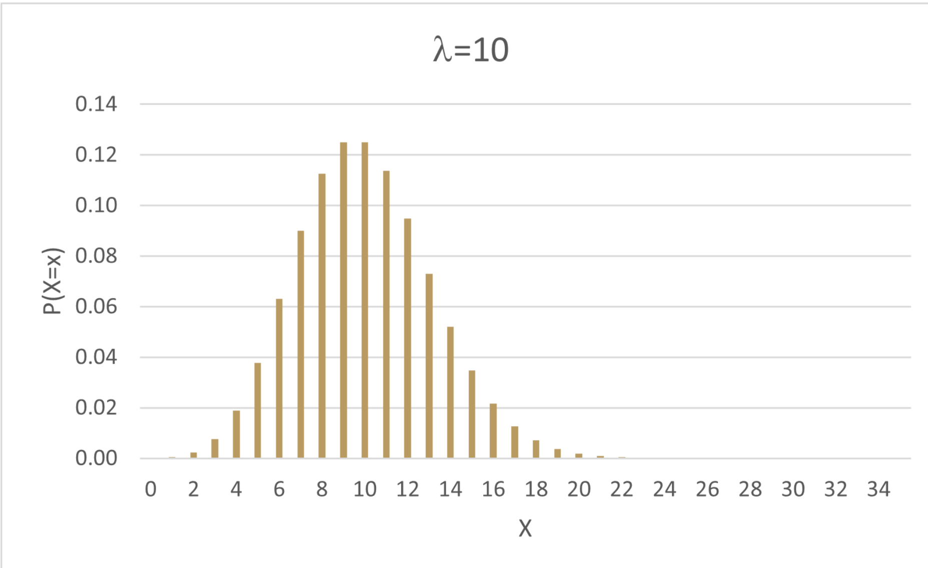
$$\mu = E(X) = \lambda$$
$$\sigma^2 = Var(X) = \lambda$$

4.4 Effect of parameters on shape of distribution

As λ increases, the distribution moves more to the right; skew to the right $\longrightarrow$ skew to the left







5 Hypergeometric distribution

Type of variable: Discrete

X = number of successes out of n observations given that the sample comes from a finite population

x = 0,1,...,n

5.1 Parameters

n = sample size

E = number of successes in population

N = population size

5.2 Probability

$\binom{n}{x} = {}_nC_x$

$$P(X = x|n, N, E) = \frac{\binom{E}{x}\binom{N-E}{n-x}}{\binom{N}{n}}$$

X	$P(X = x \mid n, N, E)$
0	$\frac{\binom{E}{0}\binom{N-E}{n-0}}{\binom{N}{n}}$
1	$\frac{\binom{E}{1}\binom{N-E}{n-1}}{\binom{N}{n}}$
$\vdots$	$\vdots$
n	$\frac{\binom{E}{n}\binom{N-E}{n-n}}{\binom{N}{n}}$

In Excel:

$P(X = x)$ =HYPGEOM.DIST(x,n,E,N,FALSE)

$P(X \leq x)$ =HYPGEOM.DIST(x,n,E,N,TRUE)

5.3 Measures

$$\mu = E(X) = \frac{nE}{N}$$

$$\sigma = \sqrt{Var(X)} = \sqrt{\frac{nE(N-E)}{N^2}}\sqrt{\frac{N-n}{N-1}}$$

6 Bivariate discrete distributions

		X			
		x_1	x_2	x_3	
Y	y_1	$P(X = x_1, Y = y_1)$	$P(X = x_2, Y = y_1)$	$P(X = x_3, Y = y_1)$	$P(Y = y_1)$
	y_2	$P(X = x_1, Y = y_2)$	$P(X = x_2, Y = y_2)$	$P(X = x_3, Y = y_2)$	$P(Y = y_2)$
	y_3	$P(X = x_1, Y = y_3)$	$P(X = x_2, Y = y_3)$	$P(X = x_3, Y = y_3)$	$P(Y = y_3)$
		$P(X = x_1)$	$P(X = x_2)$	$P(X = x_3)$	1

6.1 Calculations

6.1.1 Marginal distributions of X and Y

X	$P(X = x)$
x_1	$P(X = x_1)$
x_2	$P(X = x_2)$
x_3	$P(X = x_3)$

Y	$P(Y = y)$
y_1	$P(Y = y_1)$
y_2	$P(Y = y_2)$
y_3	$P(Y = y_3)$

6.1.2 Averages

$$
\begin{aligned}
\mu_X &= E(X) = \sum XP(X) \\
&= x_1P(X = x_1) + x_2P(X = x_2) + x_3P(X = x_3) \\
\mu_Y &= E(Y) = \sum YP(Y) \\
&= y_1P(Y = y_1) + y_2P(Y = y_2) + y_3P(Y = y_3)
\end{aligned}
$$

6.1.3 Variances

$$
\begin{aligned}
E(X^2) &= \sum X^2 P(X) \\
&= x_1^2 P(X = x_1) + x_2^2 P(X = x_2) + x_3^2 P(X = x_3) \\
Var(X) &= E(X^2) - \mu_X^2 \\
\sigma_X &= \sqrt{Var(X)} \\
E(Y^2) &= \sum Y^2 P(Y) \\
&= y_1^2 P(Y = y_1) + y_2^2 P(Y = y_2) + y_3^2 P(Y = y_3) \\
Var(Y) &= E(Y^2) - \mu_Y^2 \\
\sigma_Y &= \sqrt{Var(Y)}
\end{aligned}
$$

6.1.4 Covariance

$$
\begin{aligned}
E(XY) &= \sum_{\forall x} \sum_{\forall y} XY P(X = x, Y = y) \\
&= x_1 y_1 P(X = x_1, Y = y_1) + x_1 y_2 P(X = x_1, Y = y_2) + x_1 y_3 P(X = x_1, Y = y_3) \\
&+ x_2 y_1 P(X = x_2, Y = y_1) + x_2 y_2 P(X = x_2, Y = y_2) + x_2 y_3 P(X = x_2, Y = y_3) \\
&+ x_3 y_1 P(X = x_3, Y = y_1) + x_3 y_2 P(X = x_3, Y = y_2) + x_3 y_3 P(X = x_3, Y = y_3) \\
\sigma_{XY} &= COV(X, Y) = E(XY) - \mu_X \mu_Y
\end{aligned}
$$

6.1.5 Correlation

$$
\rho = \frac{\sigma_{XY}}{\sigma_X \sigma_Y}
$$

6.2 Proofs

$$
\begin{aligned}
\sigma_{XY} &= \sum_{i=1}^{N}\sum_{j=1}^{M}(x_i-\mu_X)(y_j-\mu_Y)P(X=x_i,Y=y_j) \\
&= E[(X-\mu_X)(Y-\mu_Y)] \\
&= E(XY - X\mu_Y - Y\mu_X + \mu_X\mu_Y) \\
&= E(XY) - E(X\mu_Y) - E(Y\mu_X) + E(\mu_X\mu_Y) \\
&= E(XY) - \mu_Y E(X) - \mu_X E(Y) + \mu_X\mu_Y \\
&= E(XY) - \mu_Y E(X) - \mu_X E(Y) + \mu_X\mu_Y \\
&= E(XY) - \mu_X\mu_Y - \mu_X\mu_Y + \mu_X\mu_Y \\
&= E(XY) - \mu_X\mu_Y
\end{aligned}
$$

$$
\begin{aligned}
E(X+Y) &= E(X) + E(Y) \\
&= \mu_X + \mu_Y
\end{aligned}
$$

$$
\begin{aligned}
[E(X+Y)]^2 &= (\mu_X+\mu_Y)^2 \\
&= (\mu_X+\mu_Y)(\mu_X+\mu_Y) \\
&= \mu_X^2 + \mu_X\mu_Y + \mu_Y\mu_X + \mu_Y^2 \\
&= \mu_X^2 + 2\mu_X\mu_Y + \mu_Y^2
\end{aligned}
$$

$$
\begin{aligned}
E[(X+Y)^2] &= E[(X+Y)(X+Y)] \\
&= E(X^2 + XY + YX + Y^2) \\
&= E(X^2 + 2XY + Y^2) \\
&= E(X^2) + E(2XY) + E(Y^2) \\
&= E(X^2) + 2E(XY) + E(Y^2)
\end{aligned}
$$

$$
\begin{aligned}
Var(X+Y) &= E[(X+Y)^2] - [E(X+Y)]^2 \\
&= E(X^2) + 2E(XY) + E(Y^2) - (\mu_X^2 + 2\mu_X\mu_Y + \mu_Y^2) \\
&= E(X^2) + 2E(XY) + E(Y^2) - \mu_X^2 - 2\mu_X\mu_Y - \mu_Y^2 \\
&= [E(X^2) - \mu_X^2] + [E(Y^2) - \mu_Y^2] + 2[E(XY) - \mu_X\mu_Y] \\
&= Var(X) + Var(Y) + 2COV(X,Y)
\end{aligned}
$$

$$
\begin{aligned}
E(aX+bY) &= E(aX) + E(bY) \\
&= aE(X) + bE(Y) \\
&= a\mu_X + b\mu_Y
\end{aligned}
$$

$$
\begin{aligned}
[E(aX+bY)]^2 &= (a\mu_X + b\mu_Y)^2 \\
&= (a\mu_X + b\mu_Y)(a\mu_X + b\mu_Y) \\
&= a^2\mu_X^2 + a\mu_X b\mu_Y + b\mu_Y a\mu_X + b^2\mu_Y^2 \\
&= a^2\mu_X^2 + 2ab\mu_X\mu_Y + b^2\mu_Y^2
\end{aligned}
$$

$$
\begin{aligned}
E[(aX+bY)^2] &= E[(aX+bY)(aX+bY)] \\
&= E(a^2X^2 + aXbY + bYaX + b^2Y^2) \\
&= E(a^2X^2 + 2abXY + b^2Y^2) \\
&= E(a^2X^2) + E(2abXY) + E(b^2Y^2) \\
&= a^2E(X^2) + 2abE(XY) + b^2E(Y^2)
\end{aligned}
$$

$$
\begin{aligned}
Var(aX+bY) &= E[(aX+bY)^2] - [E(aX+bY)]^2 \\
&= a^2E(X^2) + 2abE(XY) + b^2E(Y^2) - (a^2\mu_X^2 + 2ab\mu_X\mu_Y + b^2\mu_Y^2) \\
&= a^2E(X^2) + 2abE(XY) + b^2E(Y^2) - a^2\mu_X^2 - 2ab\mu_X\mu_Y - b^2\mu_Y^2) \\
&= a^2[E(X^2) - \mu_X^2] + b^2[E(Y^2) - \mu_Y^2] + 2ab[E(XY) - \mu_X\mu_Y] \\
&= a^2Var(X) + b^2Var(Y) + 2abCOV(X,Y)
\end{aligned}
$$

7 Questions

7.1 Question 1

Given here is the joint probability function associated with data obtained in a study of automobile accidents in which a child (under age 5 years) was in the car and at least one fatality occurred. Specifically, the study focused on whether or not the child survived and what type of seatbelt (if any) he or she used.

$$
X = \begin{cases} 0 & \text{if the child survived,} \\ 1 & \text{if not.} \end{cases}
$$

en

$$
Y = \begin{cases} 0 & \text{if no belt used,} \\ 1 & \text{if adult belt used,} \\ 2 & \text{if car-seat belt used} \end{cases}
$$

Notice that X is the number of fatalities per child and since children's car seats usually utilise two belts, Y is the number of seatbelts in use at the time of the accident.

	X		
Y	0	1	Total
0	0.38	0.17	0.55
1	0.14	0.02	0.16
2	0.24	0.05	0.29
Total	0.76	0.24	1

1. Calculate the average and standard deviation of the number of fatalities per child.
2. Calculate the average and standard deviation of the number of seatbelts in use at the time of the accident.
3. Are the two variables independent?
4. Describe the relationship between the number of fatalities per child and the number of seatbelts in use at the time of the accident.

7.2 Question 2

The manager of a stockroom in a factory has constructed the following probability distribution for the daily demand (number of times used) for a particular tool. It costs the factory $10 each time the tool is used. Find the mean and variance of the daily cost for use of the tool.

X	$P(X = x)$
0	0.1
1	0.5
2	0.4

7.3 Question 3

An urn contains ten marbles, of which five are green, two are blue, and three are red. Three marbles are to be drawn from the urn, one at a time without replacement. What is the probability that all three marbles drawn will be green?

7.4 Question 4

According to a recent study, 88% of retail brands use two or more social media channels for business. Consider a sample of 10 businesses is selected.

1. What is the probability that 7 businesses use two or more social media channels?
2. What is the average and standard deviation of the number of businesses that use two or more social media channels?
3. What is the probability that at least 8 businesses use two or more social media channels?

7.5 Question 5

Let X be a random variable with $P(X = x)$ given in the table below.

X	$P(X = x)$
1	0.4
2	0.3
3	0.2
4	0.1

Calculate:

1. $E(X)$
2. $E(\frac{1}{X})$
3. $E(X^2 - 1)$
4. $Var(X)$

7.6 Question 6

Customers arrive at a checkout counter in a department store according to a Poisson distribution at an average of seven per hour. During a given hour, what are the probabilities that

1. no more than three customers arrive?

2. at least two customers arrive?

3. exactly five customers arrive?

7.7 Question 7

A particular concentration of a chemical found in polluted water has been found to be lethal to 20% of the fish that are exposed to the concentration for 24 hours. Twenty fish are placed in a tank containing this concentration of chemical in water.

1. Find the probability that exactly 14 survive.

2. Find the probability that at least 10 survive.

3. Find the probability that at most 16 survive.

4. Find the mean and variance of the number that survive.

7.8 Question 8

The mean number of automobiles entering a mountain tunnel per 15 minutes is 7.5. An excessive number of cars entering the tunnel during a brief period of time produces a hazardous situation. Find the probability that the number of autos entering the tunnel during a one-minute period exceeds three.

7.9 Question 9

Five cards are dealt at random and without replacement from a standard deck of 52 cards. What is the probability that the hand contains all 4 aces if it is known that it contains at least 3 aces?